

Trading Applications of Generalized Probability Problems

Companion to *Generalized Probability Problems, Volume I*

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Abstract

This is a companion document to *Generalized Probability Problems, Volume I*. That volume develops fifteen problem-driven chapters in full: concrete problem, complete proof, generalisation, and trading applications integrated where the mathematics is genuinely relevant. Every result is proved and every numerical claim verified by simulation.

The present document has a narrower purpose. It compacts the trading applications of those chapters into a single reference, so that a reader interested in the practical use of each result can access the full set in a few pages, and then return to the main volume for the underlying mathematics.

Each section states the central result in the most compact form that lets the applications land, then develops the applications, then points to the corresponding chapter of the main volume. Nothing here replaces the full treatment; the objective is only to make the applications navigable on their own.

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How to read this document

Each chapter of this document mirrors a chapter of the main volume. The structure is fixed:

- **The result.** A compact statement of the probabilistic result — typically two to four lines. The main volume develops it in full, with proofs and generalisations.
- **Trading applications.** The main content of this document. Numerical claims are quoted with the assumptions under which they hold; the underlying calculations and simulations live in the main volume.
- **Reference.** A pointer to the corresponding chapter of *Generalized Probability Problems, Volume I*.

The reader is assumed familiar with elementary probability. No proofs appear in this document; wherever a claim requires justification, the justification is in the main volume.

1 Bertrand's Paradox: Defining the Probability Model

The result

Selecting a chord “at random” from a circle can give probabilities $\frac{1}{2}$, $\frac{1}{3}$, or $\frac{1}{4}$ that the chord exceeds the side of an inscribed equilateral triangle, depending on the construction. There is no contradiction: different constructions define different probability measures on the same space.

Trading applications

The paradox is a permanent warning against loosely specified probability statements. In quantitative finance the same issue recurs in four concrete forms.

- **Sampling frequency.** Daily, hourly, and tick-level samples of the same market are not equivalent statistical experiments. Estimated volatility, autocorrelation, and market impact all depend on the sampling scheme, and results are not transferable across frequencies without an argument.
- **Physical versus risk-neutral measure.** The statement “volatility is random” is incomplete: it must specify under which measure. Implied volatility, realised volatility, and model-implied dynamics live under different measures, and conflating them is the source of many pricing errors.
- **Order arrival specification.** Poisson, Hawkes, and general point processes describe order arrivals with different clustering and dependence structures. “Random arrivals” is not a model; it is a category of models.
- **Return definitions.** Arithmetic, logarithmic, excess, and total returns have different distributional properties. Choosing between them changes what a test is asking.

Practical rule. Before quoting a probability, specify the sample space, the random variable, the measure, and the sampling mechanism. Many apparent statistical edges vanish when the sampling procedure is changed.

Main volume: Chapter 1 (Geometric Paradoxes).

2 Records and the Constant e

The result

For n i.i.d. continuous observations, the probability that the n -th observation sets a new record is $1/n$. The expected number of records in the first n observations is the harmonic number

$$\mathbb{E}[R_n] = H_n \approx \log n + \gamma \quad (\gamma \approx 0.5772).$$

Record indicators are mutually independent, and record counts grow *logarithmically* in n .

Trading applications

Records are the natural null model against which to judge “extreme” events. The main content is quantitative: superlatives are far cheaper than intuition suggests.

- **Frequency of new highs is small under i.i.d. noise.** Under an i.i.d. continuous null, only $H_{10,000} \approx 9.79$ record daily returns are expected in forty years of daily data. If a series shows many more, that is evidence of dependence or regime shift; if it shows the expected number, each new high is not news.
- **The regime determines the record clock.** Records of i.i.d. observations grow as $\log n$; records of driftless cumulative sums grow as $2\sqrt{n/\pi}$ (six times faster at $n = 2000$); records of drifting sums grow *linearly*. A strategy’s equity curve lives in the third regime and sets highs constantly; “the fund made a new high” is baseline behaviour there, not a signal.
- **Multiple testing.** Tracking m statistics over n periods produces mH_n expected “best-ever” announcements under pure noise: with 20 metrics over 250 trading days, roughly 120 record-setting events per year, all meaningless. This is the record-count analogue of a Bonferroni correction.
- **Drawdown records.** The same logic applies to worst episodes. Under a stationary null a new maximum drawdown after a long quiet period is expected, not evidence of failure. Risk frameworks triggering reviews on “record drawdown” events should calibrate against the null hazard rather than treat every record as an anomaly.

The honest question is never *whether* a record occurred, but whether records occur *more often than the null clock for the relevant regime* predicts.

Main volume: Chapter 2 (Records and the Constant e).

3 The Secretary Problem and Optimal Stopping with Rankings

The result

Present n candidates one at a time in uniformly random order; each must be accepted or rejected on the spot, and only the relative rank among candidates seen so far is observable. The optimal strategy is: reject the first n/e candidates, then accept the first one better than all previously seen. As $n \rightarrow \infty$, this succeeds with probability $1/e \approx 0.368$.

Trading applications

The secretary problem is not a literal model of trading, but it isolates a principle that governs every sequential decision under partial information: *the optimal stopping rule depends on what information is available at the moment of decision*. Variants and adaptations of the framework recur across finance.

- **Rank-only versus value-based stopping.** The classical problem uses a *time* threshold because the information is ordinal. When cardinal values are observable — the actual size of an opportunity, the actual price of a quote — the threshold moves to the *value* axis. The two regimes require different tools: time cutoffs from the rank problem, price cutoffs from the value problem, both derived by Bellman recursions.
- **Execution decisions.** Whether to execute now or wait for a better quote is a stopping problem where the state contains price, spread, time remaining, and often inventory. The secretary framework does not solve it, but supplies the mental model: irrevocable decisions under a horizon require a policy, not a heuristic.
- **Sample-buying with paid observation.** A closer analogue: draw i.i.d. from a distribution, pay cost c per draw, choose when to stop and keep the last value. The optimal threshold is $1 - \sqrt{2c}$, and the expected payoff equals the threshold. This is directly relevant to search and negotiation problems in market making.
- **American options.** Exercise timing is an optimal stopping problem with cardinal information (the payoff of immediate exercise) versus a continuation value (the expected payoff of waiting). The secretary problem is the ordinal special case; American pricing is the cardinal generalisation through free-boundary methods.

The framework's value is that it makes explicit the tradeoff between *information*, *time*, and *irreversibility* that every trading decision navigates.

Main volume: Chapter 3 (Secretary Problem and Variants).

4 The Pólya Urn: Reinforcement and Path Dependence

The result

Start with a red and b black balls; at each step draw uniformly, return, and add c balls of the drawn colour. The proportion $X_n = R_n/(R_n + B_n)$ is a bounded martingale converging a.s. to a *random* limit

$$X_\infty \sim \text{Beta}\left(\frac{a}{c}, \frac{b}{c}\right).$$

The reinforcement parameter c is a dial: $c \rightarrow 0$ gives sampling with replacement (deterministic limit); $c \rightarrow \infty$ gives a coin flip between the two extremes (the first draw decides); $c = -1$ gives sampling without replacement, whose fluctuations converge to a Brownian bridge.

Trading applications

The urn is a stylised model of processes where past observations alter future probabilities. Its practical value is as a warning and as a generator of intuition.

- **Reinforcement in markets.** Trades that move prices attract attention and further trades; early adopters shape consensus; strategies that visibly work draw capital that reinforces their moves. The urn is not literal here, but its message is exact: even with no drift, feedback makes the long-run outcome *random*, not deterministic.
- **Bayesian updating in production.** Any incrementally updated estimator of a binary probability — a fill rate, a signal hit ratio, a crossing probability — has the exact structure of a Pólya urn with specific parameters. The initial pseudo-counts (a, b) encode the prior belief; the increment c determines how fast the estimator reacts to new observations. This is not just a modelling choice: the same asymptotic variance formula $\text{Var}(X_\infty) = abc/((a+b)^2(a+b+c))$ shows that the parameter c sets the residual randomness of the long-run estimate. Choosing the reinforcement is choosing the sensitivity of the estimator for the entire lifetime of the strategy.
- **Path dependence in market microstructure.** Two intervals with identical aggregate statistics can have different predictive properties if their histories differ. Concrete cases: two five-minute windows with the same net order flow but different arrival profiles (continuous versus clumped) have different adverse-selection risk; two sessions with the same realised volatility but different volatility paths have different implied surfaces; two identical-looking order books can have different effective depth depending on whether they were built by fresh liquidity or reconstructed after cancellations. In all three the mechanism is reinforcement — trades attract trades, volatility begets volatility, cancellations signal intention. Any model asserting that the current state contains all predictive information is denying the presence of this mechanism.
- **The card-deck analogue.** Sampling without replacement from a deck of $N/2$ reds and $N/2$ blacks has imbalance converging (rescaled) to a Brownian bridge. This is the continuous form of “the deck remembers what has been drawn,” and it underlies the red-black card stopping problem below — and, in a genuine trading application, any finite-inventory model where remaining supply constrains future prices.

The urn’s practical lesson is that reinforcement is not a rare pathology. It is present

wherever attention, capital, or memory affects future probabilities, and it changes what independence-based statistics mean.

Main volume: Chapter 4 (Urn Models and the Pólya Urn).

5 The Egg-Drop Problem: Sequential Design under Resource Constraints

The result

With k identical eggs and F floors, and a threshold floor f^* above which any drop breaks the egg, the number of floors resolvable with n drops and k eggs is

$$f(n, k) = \sum_{i=1}^k \binom{n}{i}.$$

The recursion $f(n, k) = f(n-1, k-1) + 1 + f(n-1, k)$ captures the two possible outcomes of any drop.

Trading applications

The egg-drop problem has no direct trading analogue, but its structure — optimal sequential experimentation under a finite budget — is recurrent. The applications are methodological.

- **Sequential parameter search and overfitting.** When tuning a strategy on limited data, each backtest consumes statistical power that no additional data recovers. Testing k parameter configurations on the same sample inflates the probability of finding a spurious winner in exactly the multiple-testing sense; the egg-drop analogy is literal, with each backtest playing the role of a drop from a chosen floor. The design question — which parameter to test next, given what previous tests have revealed — is a sequential-experimentation problem with the same recursive structure as the classical egg-drop recursion, and the reason to formalise it is precisely to avoid the overfitting that ad-hoc parameter sweeps guarantee.
- **Adaptive execution as sequential experimentation.** An adaptive VWAP schedule (one that adjusts child-order sizes based on the impact observed from previous ones) faces the same structural tension as the egg-drop problem: each child order both *executes* — consuming part of the parent-order budget — and *probes* — revealing information about the market's current impact function. The two roles cannot be separated: probing requires trading, and trading commits capital. Sizing the next child order aggressively learns quickly about impact but risks depleting the schedule if the guess is wrong; sizing conservatively preserves optionality but leaves the rest of the schedule under-informed. The optimal policy — explore aggressively early, exploit later when uncertainty is reduced — has the same recursive shape as the egg-drop recursion, with the residual quantity to execute and time remaining playing the roles of remaining eggs and floors.
- **Cost-constrained testing.** Whenever an experiment moves the market (or moves the counterparty, or moves the model), each observation carries a cost that alters subsequent conditions. Optimal design under such constraints is the domain of the same recursion.

The value of the problem is that it forces the researcher to formalise what “budget” means in a given setting, and to compute the value of each experiment against what it consumes.

6 The Red–Black Card Problem: Stopping under Deck Memory

The result

Draw sequentially from a shuffled deck of 26 red and 26 black cards without replacement; the score after k draws is $S_k = (\text{reds}) - (\text{blacks})$; stop and receive S_k at any time. The game has *positive* value: with $N = 52$ the optimal expected payoff is approximately 2.62. The mechanism is that (S_k) is not a martingale — its conditional drift $-S_k/(N - k)$ pulls it back toward zero — so waiting when $S_k < 0$ is favourable, and stopping when $S_k > 0$ locks in a lead that would otherwise erode. The optimal strategy is a Bellman recursion.

Trading applications

The card problem is the cleanest illustration of stopping under finite-population dynamics, and its intuition transfers to any setting where remaining supply is constrained.

- **Stop when winning, hold when losing — with a reason.** The value comes from an asymmetry: the deck owes you what has not been drawn. When ahead, the remainder is unfavourable and the drift erodes your lead; when behind, the remainder is favourable and the drift restores it. This is the discrete counterpart of the Brownian bridge's mean reversion, and the intuition is exact.
- **Finite-inventory execution.** An informed trader working a large parent order faces a bounded remaining quantity, with each executed slice reducing the counterparty pool. The setting is not identical to the card deck, but the combinatorial structure is: what happens next depends on what has already been observed, and stopping rules must incorporate this dependence.
- **Where a naive martingale argument fails.** If one treats S_k as a martingale, optional stopping gives value zero and no strategy profits. That reasoning applies only *with* replacement. The failure of the martingale shortcut without replacement is a general warning: many trading problems are stated in i.i.d. language but happen in finite populations, and the finite-population correction changes both the sign and the size of the answer.
- **Why Bellman is unavoidable.** The optimal strategy cannot be summarised by a simple threshold on S_k alone; it depends on the joint state (current score, remaining cards). Dynamic programming is not a stylistic choice but a structural necessity, and the same is true of most trading problems that involve inventory, time, and an option to act.

Main volume: Chapter 6 (Red–Black Card Stopping).

7 Pattern Hitting Times: Guibas–Odlyzko and Self-Overlap

The result

For a fair coin and any pattern P of length n , the expected time to first observe P as a contiguous block is

$$\mathbb{E}[T_P] = \sum_{i=0}^{n-1} c_i(P) 2^{n-i},$$

where $c_i(P)$ is the pattern’s autocorrelation vector: $c_i = 1$ if the length- $(n-i)$ prefix of P equals its length- $(n-i)$ suffix, zero otherwise. Two patterns of the same length can have very different expected waiting times: HHH takes 14 tosses on average, HTH takes 10, TTH takes 8. The difference is the pattern’s self-overlap.

Trading applications

The result matters because pattern searches in market data are ubiquitous, and the naive assumption that patterns of the same length are equivalent is wrong.

- **Pattern mining and rare-looking sequences.** Traders and researchers routinely scan sequences of {buy, sell}, {up, down}, or discretised volatility states for “significant” patterns. Guibas–Odlyzko shows that patterns with heavy self-overlap are structurally slow to occur; a pattern appearing more often than baseline is unusual only if measured against *that pattern’s* expected waiting time, not a generic 2^n .
- **Sequential tests and stopping-time bias.** If a test stops the first time a pattern appears, the resulting sample is not a uniform slice of the process. This is closely related to the multiple-testing inflation of Chapter 2: conditioning on the first occurrence of a rare event changes the distribution of what one has seen up to that point.
- **Conway’s leading-number algorithm for pattern races.** For a race between two patterns, the winner and its expected time to victory are given by a simple algorithm on their autocorrelation structure. Applications include order-flow diagnostics: which of two market states arrives first, on average, given the current dynamics.
- **Martingale reasoning and the “infinite casino” argument.** The Guibas–Odlyzko theorem admits a proof via a population of independent gamblers betting on the pattern appearing at each step, with optional stopping killing the sum of stakes. This construction of a martingale that absorbs a stopping problem is a recurring technique in derivative pricing — the same shape appears in risk-neutral valuation, where the discounted price process is a martingale under the pricing measure.

Main volume: Chapter 7 (Pattern Hitting Times).

8 Order Statistics: Maxima, Minima, and the Max–Min Toolkit

The result

For any random variables X, Y ,

$$\max(X, Y) + \min(X, Y) = X + Y, \quad \max(X, Y) - \min(X, Y) = |X - Y|.$$

The second identity reduces the expected absolute difference $\mathbb{E}|X - Y|$ to a difference of order-statistic means. For n i.i.d. samples from $\text{Uniform}(0, S)$,

$$\mathbb{E}[U_{(k)}] = \frac{k}{n+1}S,$$

so in particular $\mathbb{E}[\max] = \frac{n}{n+1}S$ and $\mathbb{E}[\min] = \frac{1}{n+1}S$.

Trading applications

Order statistics show up whenever a system depends on the best, worst, first, or last observation, rather than the average one. Four settings are direct.

- **Range-based volatility estimation.** Parkinson's estimator uses the daily high–low range as a variance estimator with roughly five times the efficiency of the close-to-close squared return. The high and low are the max and min of the intraday path, and the identity $\mathbb{E}[(\max - \min)^2]$ is the continuous limit of the discrete identities above.
- **Empirical quantiles and historical VaR.** The k -th order statistic *is* the empirical quantile: a historical 99% VaR from $n = 500$ daily returns is the 5-th order statistic of the losses. The formula $\mathbb{E}[U_{(k)}] = k/(n+1)$ explains both why the estimator is centred correctly and why it is noisy near the tail: order statistics are spaced $\sim 1/(n+1)$ apart in probability but far apart in value where the density is low.
- **Auction revenue equivalence.** In a second-price auction with n i.i.d. uniform valuations, expected revenue is $\mathbb{E}[U_{(n-1)}] = (n-1)/(n+1)$; in a first-price auction under symmetric equilibrium, the winner bids $b(v) = (n-1)v/n$ and expected revenue is the same. Both formats reduce to the same order-statistic mean — the Vickrey–Myerson revenue-equivalence theorem in its simplest form.
- **Portfolio stop-outs as minimum hitting times.** If a portfolio is liquidated when the *first* of several open positions hits its stop, the effective portfolio hitting time is $\min(T_1, \dots, T_n)$, the minimum of the individual hitting times. The consequence is asymmetric: the probability that a portfolio stops out in a given window grows with the number of positions, and does so much faster than the individual probability. A portfolio of n positions each with 10% daily stop probability has roughly $1 - 0.9^n$ portfolio-level stop probability (40% at $n = 5$, 65% at $n = 10$) if the positions are independent, and worse if drawdowns are correlated. Any risk framework built on individual stop probabilities without an order-statistic correction will systematically understate portfolio-level stop frequency.
- **Dispersion of execution prices across sliced orders.** If a large parent order is worked as n child fills, and the fill prices are modelled as i.i.d. samples from a band of width R around the prevailing level, the expected range $\mathbb{E}[\max] - \mathbb{E}[\min]$ of the fills grows toward R as n increases. This is a statement about the *dispersion* of realised

prices across the child fills, not about slippage against a benchmark: finer slicing reduces market impact per child but widens the observed range of realised prices. The two quantities are distinct and often traded off against each other in execution algorithm design.

Main volume: Chapter 8 (Order Statistics).

9 Size-Biasing: When Sampling by Encounter Lies

The result

If i.i.d. intervals X have mean μ and one observes the interval containing a uniformly random instant, the expected length of that interval is not μ but the *size-biased* mean

$$\frac{\mathbb{E}[X^2]}{\mathbb{E}[X]} = \mu(1 + CV^2), \quad CV = \sigma/\mu.$$

The residual waiting time until the next event is $\mathbb{E}[W] = \mathbb{E}[X^2]/(2\mathbb{E}[X])$. The bias vanishes only when the distribution is degenerate; the more variable the intervals, the larger the gap.

Trading applications

Size bias is present whenever the sampling mechanism is “encounter during a period” rather than “uniform over events.” Financial data is riddled with such sampling.

- **Inter-trade times and passive quotes.** A market observed at random calendar instants oversamples long quiet periods. Estimating the expected time a passive quote waits by averaging inter-trade intervals from calendar-time snapshots systematically overstates the wait, and by CV^2 times more when trade activity is bursty.
- **Event-time versus calendar-time sampling.** The two natural clocks for market data give systematically different estimates of “typical” volatility, and the difference is exactly a size-biasing effect. In *calendar time*, each interval has fixed duration; the many quiet hours in a normal history dominate the sample, and volatility estimates *understate* the impact of active episodes. In *event time* (fixed number of trades, fixed volume, fixed tick count per interval), each interval has fixed activity but variable duration; active episodes now dominate the sample because they generate more events per unit of time, and volatility estimates *overstate* typical conditions. The two estimators are not wrong in any absolute sense — they measure different things — but their difference is not a nuisance to average away. It is a real feature of the joint distribution of activity and price change, and any calibration that mixes the two clocks silently is arithmetically inconsistent.
- **Snapshot bias in position durations.** A snapshot of currently open positions over-represents strategies whose trades stay open a long time. Reporting “average holding period” from live inventory is a length-biased estimate; the population-average holding period, computed from the full history of opened and closed trades, is smaller by the same factor.
- **Credit and default event timing.** If default events are separated by heterogeneous intervals, an observation at a random date lies inside a longer-than-average gap, and the expected residual time to the next default is not half the unconditional mean but $\mathbb{E}[X^2]/(2\mu)$. Reporting “time since last default” as an intuition for “time until next” is exactly the bus paradox.

The general rule is that sampling a process in calendar time is not equivalent to sampling its events uniformly, and any estimator built from the wrong clock inherits a bias proportional to CV^2 .

Main volume: Chapter 10 (Size-Biasing and Renewal Theory).

10 Random Walks and Gambler's Ruin

The result

For a simple symmetric walk on $\{0, 1, \dots, N\}$ starting at k , absorbed at the endpoints, the probability of reaching N before 0 is

$$p_k = \frac{k}{N},$$

and the expected time to absorption is $T_k = k(N - k)$. For an asymmetric walk with $p = \mathbb{P}(\Delta = +1)$, $q = 1 - p$, $r = q/p$,

$$p_k = \frac{1 - r^k}{1 - r^N}.$$

Both quantities are solutions of *discrete Poisson equations*: the hitting probability is harmonic (source zero), giving a linear profile; the expected time has a constant source, giving a parabolic profile. The framework generalises verbatim to Brownian motion.

Trading applications

The classical gambler's ruin is not a market model, but its structure appears in every risk framework that has an absorbing boundary.

- **Ruin under positive edge.** A strategy with positive expected return can still hit a hard drawdown boundary if leverage is high enough. The asymmetric formula makes this precise: even at $p > q$, p_k can be small when k is close to the ruin boundary relative to N . Positive expectation does not preclude ruin; *distance from the boundary does*, and leverage rescales that distance directly. The Kelly criterion (Chapter 15) is the answer to the question this raises — given an edge μ and a volatility σ , at what leverage is the ruin probability tolerable relative to the growth achieved. Kelly and gambler's ruin are the same decision viewed from two angles: one asks how far from the boundary you should stand, the other asks how large a step you should take.
- **Leverage as distance rescaling.** Doubling position size halves the effective distance to a stop loss, measured in units of unit-position P&L increments. Ruin probability and expected time to hit therefore depend on leverage through this rescaling rather than on any direct dependence on position size.
- **Barrier options as first-passage problems.** Pricing knock-in and knock-out options is the continuous analogue of computing p_k and T_k for the random walk. The reflection principle (Chapter 11) is the tool that makes the computations tractable in closed form under Brownian motion.
- **Absorbing states in capital management.** Once capital hits a forced-liquidation boundary — margin call, fund gate, redemption cascade — the process cannot recover on its own. This makes p_k (the probability of reaching the boundary) more decisive than expected P&L. Every position-sizing rule is implicitly a choice about which p_k is acceptable.

Main volume: Chapter 11 (Random Walks and Gambler's Ruin).

11 The Reflection Principle and the Ballot Problem

The result

The reflection principle counts constrained lattice paths by mapping them bijectively to unconstrained ones: paths from a to b that touch a barrier c are in bijection with paths from the reflected starting point $2c - a$ to b . As a canonical corollary, the ballot problem: if candidate A receives p votes and B receives $q < p$, then under a uniformly random counting order the probability that A leads throughout is

$$\frac{p - q}{p + q}.$$

Trading applications

Reflection is the combinatorial engine behind every first-passage computation for random walks and their continuous limit, Brownian motion. Its direct applications are in path-dependent pricing and in risk analysis of trajectories that must cross thresholds.

- **Barrier option pricing.** Knock-in, knock-out, and rebate options depend on whether the underlying touches a level before expiry. The reflection principle for Brownian motion — the direct continuous analogue of the discrete bijection — provides the closed-form distributions of maxima and hitting times used in these prices.
- **Maximum drawdown distribution.** The joint law of a random walk and its running maximum is the object of the reflection principle. Drawdown limits, calibration of risk-of-ruin figures, and the timing of drawdowns (linked to the arcsine law of Chapter 13) all sit in this framework.
- **Liquidation and margin thresholds.** The probability that an asset breaches a margin threshold before exiting at a target level is a first-passage question with two absorbing barriers, and it is a reflection computation. The same object arises in stop-loss versus take-profit calibration: which is hit first, and with what expected time.
- **The ballot problem as a template.** The formula $(p - q)/(p + q)$ answers a specific question: given a predetermined final tally with A leading by $p - q$, what is the probability that A led throughout the counting? The result is not about the winner — that is fixed — but about the *path* to victory. This structure appears in trading whenever the aggregate outcome is known but the temporal dominance is not:
 - Two venues each receiving fills of a parent order — how often does the eventual winner lead throughout the working day, versus catching up late? A venue that dominates consistently versus one that alternates gives different diagnostic signals for routing.
 - Two strategies compared on the same period — if A ends with more winning trades than B , the probability that A led throughout, under random ordering of the outcomes, calibrates whether the aggregate superiority reflects genuine dominance or a late reversal.
 - Buy versus sell volume accumulating in a session — given the final imbalance, the probability of persistent dominance by one side against alternation is a null baseline for order-flow diagnostics.

In each case the interest is the path, not the endpoint, and $(p - q)/(p + q)$ is the null against which observed persistence is judged.

The general lesson is that any question of the form “does the path touch this level before that one” has a reflection-principle answer, and the combinatorial object is the same across problems.

Main volume: Chapter 12 (The Reflection Principle and the Ballot Problem).

12 First Crossing of the Majority: Catalan Numbers and $\pi/4$

The result

Toss a fair coin repeatedly. Stop the first time the running proportion of heads exceeds $\frac{1}{2}$; record the final proportion a . Under the reduction $S_n = 2H_n - n$, this is the first passage of a simple symmetric walk to $+1$, and the first-passage distribution is governed by the Catalan numbers. The expected final proportion is

$$\mathbb{E}[a] = \frac{\pi}{4} \approx 0.7854.$$

The generating-function computation — summing Catalan-weighted central-binomial coefficients against $1/(2k+1)$ — passes through an integrated arcsine, and π enters via $\arcsin(1) = \pi/2$. For a biased coin with $\mathbb{P}(H) = p \geq \frac{1}{2}$, the same mechanism yields the closed form $\mathbb{E}[a] = p \arcsin(2\sqrt{pq})/(2\sqrt{pq})$.

Trading applications

This chapter has no direct trading applications of its own: the value of $\pi/4$ is remarkable as a demonstration of how π emerges from coin tosses, but the practical uses of the underlying machinery — the central-binomial kernel $1/\sqrt{1-4x}$, the arcsine integration, and the walk-based reduction — are developed in the following chapter. The two problems share a common analytic source: the same kernel that produces $\pi/4$ here produces the arcsine density there.

The methodological point worth carrying forward is the reduction step itself: the seemingly analytic condition “proportion exceeds one half” becomes, after the change of variables $S_n = 2H_n - n$, the purely combinatorial condition “the walk reaches $+1$.” Recognising proportion thresholds as random-walk level crossings is a recurring move in derivative pricing and in the analysis of strategy statistics — the same idea underlies the ballot problem, the red–black card stopping problem, and the arcsine law of the next chapter.

Main volume: Chapter 13 (First Crossing of the Majority). Applications developed in Chapter 14.

13 The Arcsine Law for the Time Spent Positive

The result

Toss a fair coin $2n$ times; let L_{2n} be the number of steps the walk S_k spends on the positive side. Lévy's discrete arcsine law gives $\mathbb{P}(L_{2n} = 2k) = u_{2k} u_{2n-2k}$, where $u_{2j} = \binom{2j}{j}/4^j$ is the return-to-origin probability. In the continuous limit,

$$\frac{L_{2n}}{2n} \rightarrow X, \quad f_X(x) = \frac{1}{\pi\sqrt{x(1-x)}}, \quad F_X(x) = \frac{2}{\pi} \arcsin \sqrt{x}.$$

The density is *U-shaped*: minimum at $x = 1/2$, divergent at 0 and 1. Extreme splits are far more likely than balanced ones. A companion arcsine law governs the location of the running maximum: $\tau_{\max}/T$ is also arcsine-distributed on $[0, 1]$.

Trading applications

The arcsine law is the sharpest quantitative correction to the naive intuition that random walks oscillate symmetrically around zero. Its practical implications are counterintuitive and important.

- **P&L paths streak, not oscillate.** A driftless cumulative P&L spends, in law, most of its time on one side of breakeven. Extended runs above or below water are expected features, not evidence of drift. Reading such a path as “the strategy is trending” when it has no edge is the classic error the arcsine law diagnoses.
- **Time above high-water mark is a poor edge signal.** A zero-edge manager will show, on any finite horizon, a record dominated by extended above-water or below-water streaks. The fraction of time an equity curve was above its high-water mark is arcsine-distributed and can take arbitrarily high or low values by chance. Concretely, from the CDF: a 6.4% probability of spending less than 1% of the time positive, and by symmetry another 6.4% of spending more than 99% of the time positive. Nearly 13% of zero-edge strategies will look almost entirely one-sided over any finite horizon — extreme fractions are not exceptional under the null, they are the modal outcome.
- **Last-crossing intuition is inverted.** The last time the walk visited zero before T is itself arcsine-distributed on $[0, T]$. A spread that has not touched its fair value for a long time is *not* thereby more likely to revert soon; long absences are common. The naive “it is due for a reversal” intuition inverts the truth — conditional on a long absence, the expected time to the next visit grows.
- **The timing of maximum drawdowns is not uniform.** A companion arcsine law governs the location of the running maximum: for a walk observed over $[0, T]$, the instant $\tau_{\max}$ at which the maximum is achieved is arcsine-distributed on $[0, T]$. The density is U-shaped on $[0, T]$, mode-less, and *minimised* at $T/2$. Concretely, the probability that the maximum lies in the first or last 5% of the window is roughly 29%, against the 10% that uniform intuition suggests.

The trading consequence follows from the fact that the worst drawdown of a strategy is anchored to the location of its global peak. If the peak sits near the start of the observation window, the drawdown has developed over most of the history — a persistent, chronic drawdown. If it sits near the end, the drawdown is fresh, ongoing,

and by definition has had no time to recover. Both scenarios — old-peak chronic drawdowns and new-peak fresh drawdowns — are systematically more probable than mid-window drawdowns. A risk framework that assumes drawdown extremes fall uniformly through the observation period therefore understates two distinct risks: new-strategy risk (short windows concentrate their peak near the start), and end-of-sample risk (long windows concentrate their peak near the end). Neither is a nuisance of small samples; both are structural features of the trajectory of any driftless cumulative process.

The unifying trading lesson is that the natural mean-reversion of independent draws is not present in cumulative paths: their extremes cluster in time, on one side, and near the endpoints of the sample.

Main volume: Chapter 14 (The Arcsine Law).

14 The Longest Run of Heads: Extreme Values on Discrete Sequences

The result

The longest run of heads L_n in n fair tosses satisfies Schilling's asymptotic

$$\mathbb{E}[L_n] \approx \log_2 n + \frac{\gamma}{\ln 2} - \frac{3}{2} \approx \log_2 n - 0.667,$$

accurate to a tenth of a unit at practical sample sizes. The generalisation to a biased coin, with $\mathbb{P}(H) = p$, is

$$\mathbb{E}[L_n] \approx \log_{1/p}(n(1-p)) + \frac{\gamma}{\ln(1/p)} - \frac{1}{2}.$$

The longest run grows *logarithmically* in n , only mildly sensitive to p , because the base of the logarithm moves slowly.

Trading applications

The longest-run problem is the sharpest null model for evaluating streaks in financial data — losing runs, patterns in order flow, consecutive P&L signs.

- **Maximum losing streak in a backtest.** For a strategy whose daily P&L is a fair coin, the expected longest losing streak over one trading year ($n = 252$) is about 7.3 consecutive losses; over five years, about 9.6. A stop-loss rule that halts trading after k consecutive losing days will trigger by chance — with no strategy failure — for any k near $\log_2 n$. The right-tail decay is exponential: each unit above the expected streak roughly halves the probability of triggering, so a stop set at $k = \mathbb{E}[L_n] + 3$ triggers under the null with probability around 1/8 per year, and at $\mathbb{E}[L_n] + 5$ with probability around 1/32. Rules calibrated fewer than three units above the expected streak will fire regularly on strategies with no actual failure.
- **Multiple-testing inflation across strategies.** Backtesting m strategies over n days and examining the worst streak among them is the pooled version of the same problem, and the expected worst-case streak is $\log_2(mn) - 0.667$. For 30 strategies over one year: about 12.2 consecutive losses expected *under pure noise*. For 100 strategies over five years: about 16.3. Reading such a streak as “strategy failure” is a multiple-testing error — the same shape as a Bonferroni correction, on streak lengths rather than p -values.
- **Order-flow pattern detection.** A run of 12 consecutive buys among 4000 trades sits within $\log_2 4000 - 0.667 \approx 11.3$: baseline behaviour, not news. Genuinely anomalous runs are those exceeding the baseline by several units, each unit being exponentially less likely.
- **Edge affects frequency more than magnitude.** A strategy with a 10% edge ($q = 0.45$ instead of 0.5) reduces the expected worst yearly losing streak from 7.3 to 6.4 — less than one trade. What edge sharply reduces is the *frequency* with which long streaks occur, not their typical length. Concretely: the expected number of losing streaks of length ≥ 10 in a year of $n = 252$ trades goes from $n \cdot 0.5^{10} \approx 0.25$ under the null to $n \cdot 0.45^{10} \approx 0.09$ under the 10% edge — a factor of nearly three, against only a 15% change in expected worst length. The asymmetry has a structural reason: length

is logarithmic in n and moves slowly with q ; frequency of long streaks is polynomial in q and moves fast. The operational consequence is that stop-loss rules should be calibrated to the realised q , and that even a strategy with real edge should budget psychologically and financially for six consecutive losses in a typical year — what a positive edge buys is not the absence of long streaks, but their rarity.

The compact rule: streaks scale as $\log_2(mn)$ under the null, are insensitive to modest edges, and long runs by themselves are almost never evidence of anything except sample size and the number of things being watched.

Main volume: Chapter 15 (Longest Run of Heads).

15 The Kelly Criterion and Volatility Drag

The result

Betting a fixed fraction f of capital on an i.i.d. opportunity, the almost-sure long-run growth rate is

$$g(f) = \mathbb{E}[\log(1 + fR)] \approx f\mu - \frac{f^2\sigma^2}{2},$$

maximised at the Kelly fraction $f^* = \mu/\sigma^2$ with maximum

$$g^* = \frac{\mu^2}{2\sigma^2} = \frac{\text{Sharpe}^2}{2}.$$

The growth parabola $g(f) = g^* - (\sigma^2/2)(f - f^*)^2$ vanishes at $f = 0$ and at $f = 2f^*$; beyond twice Kelly the growth rate turns negative. For an even-odds binary bet the growth rate equals $\log 2 - H(p)$, the mutual information between belief and outcome; the general form $D(p\|r)$ (Kullback–Leibler divergence between subjective probabilities and market-implied ones) is the growth rate achievable by proportional betting.

Trading applications

Kelly is the mathematical bedrock of position sizing, and the details of its behaviour — the asymmetry between over- and under-betting, the meaning of fractional Kelly, the identification of growth with information — all matter in practice.

- **Why expectation-maximisation fails.** A strategy maximising $\mathbb{E}[W_n]$ bets everything and goes bankrupt with probability one. The expectation is dominated by exponentially rare explosion paths that no real capital ever traverses. Kelly maximises $\mathbb{E}[\log W_n]$ instead, and the strong law of large numbers guarantees the growth rate on nearly every trajectory.
- **Volatility drag as a tax.** The geometric growth of a compounded return is $g \approx \mu - \sigma^2/2$: the arithmetic mean minus half the variance. Doubling position size doubles μ but quadruples the drag, and the asymmetry is the entire content of position sizing. Two portfolios with the same arithmetic return but different volatilities have different long-run wealth outcomes.
- **Growth rate equals Sharpe²/2.** The optimal growth rate depends on μ and σ only through their ratio. Two strategies with the same Sharpe compound at the same optimal rate; they differ only in how much leverage gets each of them there. A strategy with $\mu = 20\%$, $\sigma = 20\%$ (Sharpe = 1) reaches its optimum at $f^* = 5$; one with $\mu = 5\%$, $\sigma = 5\%$ (same Sharpe) requires $f^* = 20$. Sharpe measures achievable growth; the leverage needed to extract it measures the operational cost — balance sheet, financing, capacity. Two strategies can be equivalent in expectation but very different in what they demand from the firm.
- **The asymmetry of mis-sizing.** The parabola is flat at the top and steep on the sides. Underbetting slightly loses a little growth; overbetting slightly loses more, and overbetting substantially can destroy the strategy. Given that μ and σ are always estimated, the practical asymmetry is decisive: professional sizing is deliberately below full Kelly, not above.

- **Fractional Kelly as a margin of error.** Betting cf^* with $c < 1$ reduces the growth rate to $c(2 - c)g^*$: at $c = 1/2$, three quarters of optimal growth is preserved. But drawdown probabilities decrease *much* more sharply. Under full Kelly the probability of the capital reaching $1/a$ of the current level before doubling is (q/p) ; under fractional Kelly at level c this probability becomes $(q/p)^{1/c}$. Half Kelly therefore *squares* every such probability, and quarter Kelly raises them to the fourth power. On a 60/40 coin, the probability of halving before doubling goes from 67% at full Kelly to 44% at half and 20% at quarter — for a growth cost of one quarter and nine sixteenths respectively. Half Kelly is not timidity; it is a rational hedge against overestimating the edge.
- **Kelly and losing streaks.** A Kelly bettor must survive the streaks the previous chapter quantifies. A losing streak of length $\log_2 n$ at fraction f multiplies capital by $(1 - f)^{\log_2 n}$. At full Kelly on a 60/40 coin ($f^* = 0.2$), an expected yearly streak of length 7.3 produces a drawdown of about 80% — correctly sized, and still psychologically insurmountable. At half Kelly ($f = 0.1$), the same streak produces a drawdown of about 52%; at quarter Kelly ($f = 0.05$), about 30%. Fractional sizing converts an insurmountable drawdown into a discouraging one, and sizing decisions are ultimately drawdown-tolerance decisions.
- **Kelly and information.** The growth rate equals the mutual information between belief and outcome, or, more generally, the KL divergence between subjective and market-implied probabilities. *Your growth rate is the distance between your beliefs and the price.* If the market agrees with you — $r = p$ — there is no edge, however confident you are. Edges are information, and compounding is the mechanism that pays for them.
- **Kelly as an upper bound in practice.** Real edges are estimated, not known, and Kelly is derived under the assumption of a known distribution. Half Kelly, quarter Kelly, or explicitly conservative sizes are the operational answer: Kelly sets the ceiling of rational size, and professional sizing lives below it by a margin proportional to estimation uncertainty.

Main volume: Chapter 16 (The Kelly Criterion).

16 Background: Where to Find the Foundations

This document has treated fifteen problems in the form “result plus trading applications,” with the derivations left to the main volume. A reader wanting to work through the mathematical background will find it in the appendix to *Generalized Probability Problems, Volume I*, which collects nine entries used across the chapters above. The entries are:

- **Stirling’s approximation and the harmonic numbers**, which underlie the asymptotics in Chapter 2 (records), Chapter 13 (arcsine), and Chapter 15 (longest run).
- **The Poisson paradigm**, the source of the approximation used in Chapter 15.
- **The Beta family as a unifier**, connecting order statistics (Chapter 8), the Pólya urn limit (Chapter 4), and the arcsine law (Chapter 13, which is $\text{Beta}(1/2, 1/2)$).
- **The Gumbel distribution and extreme-value theory**, the continuous analogue of the discrete extremes in Chapter 15.
- **Martingales and optional stopping**, used throughout: Chapter 4 (Pólya urn), Chapter 7 (Guibas–Odlyzko), and Chapter 16 (Kelly).
- **Generating functions as an analytic toolbox**, the engine of Chapter 7, Chapter 12 (first crossing), and Chapter 13 (arcsine).
- **de Finetti’s representation theorem**, the exchangeability result that identifies the Pólya urn’s limit and, more generally, connects rank-based and value-based models.
- **Geometric distributions and their conventions**, used in the pattern-hitting arguments of Chapter 7 and the waiting-time computations of Chapter 15.
- **Notation and conventions**, including the sign of c in the Pólya urn and the treatment of zero-crossings in the arcsine chapter.

The nine entries are stated in the format *statement, use in this volume, and reference to a standard textbook*. No proofs are given in the appendix; the results are established either in the main chapters or in the references cited there.

Main volume: Appendix (Mathematical Background).